Adaptive Multimodal Surgical Gesture Recognition via Pretrained Visual Representations and Reliability-Aware Dynamic Gating

Deepak Soni, Aparna Banpela, and Princy Matlani

Abstract—Automated surgical gesture recognition is a cornerstone of context-aware assistance, objective skill assessment, and autonomous execution in robot-assisted minimally invasive surgery (RMIS). While endoscopic video conveys semantic context of tool-tissue interactions, robot kinematics provides continuous physical trajectories immune to optical degradation. However, conventional multimodal models frequently rely on static concatenation or uncalibrated late fusion, leaving them vulnerable when visual corruptions (smoke, fog, blur) or telemetry dropouts occur. In this work, we present an empirical benchmark of an adaptive multimodal framework evaluated across all 103 trials of the JIGSAWS benchmark: Suturing (39 trials), Knot Tying (36 trials), and Needle Passing (28 trials). We combine frozen surgical foundation model features (SurgMotion-ViTL, 1024-D) and ImageNet ResNet-18 embeddings (512-D) with normalized 76-D da Vinci kinematics. Lightweight temporal sequence models (GRU) capture temporal dynamics, while a reliability-aware dynamic gating network calibrates modality confidence from representation statistics and explicit availability flags to enforce exact single-modality fallback under sensor failure. Under official Leave-One-User-Out (LOUO) cross-validation across all active folds (8 folds for Suturing and Knot Tying; 7 active folds for Needle Passing due to an empty test set for Subject G in the official distribution), empirical evaluations demonstrate: (i) Normalized 76-D kinematics provides strong unimodal gesture discrimination (66.64% accuracy on Suturing, 61.09% on Knot Tying, 48.69% on Needle Passing); (ii) Simple fixed late fusion degrades frame accuracy (-2.14% to -12.75%) by allowing visual noise to dilute confident kinematics predictions; (iii) Adaptive Reliability Gating achieves the highest accuracy across all three tasks (72.28% on Suturing, 72.34% on Knot Tying, and 51.48% on Needle Passing), demonstrating a statistically significant $+11.25\%$ gain on Knot Tying ($p = 0.0095$, Holm-Bonferroni $p_{\text{adj}} = 0.0286$, Cohen's $d_z = 1.25$); (iv) Multimodal temporal modeling substantially stabilizes sequence coherence, producing large and statistically significant improvements in segmental Edit Score ($+23.46\%$ to $+34.26\%$, $p < 0.001$, effect sizes $d_z > 2.7$); and (v) Under complete optical telemetry failure, the adaptive gate dynamically routes prediction mass to kinematics, preserving performance with minimal degradation ($\leq 0.42\%$ drop on Suturing and Needle Passing).

Index Terms—Surgical Gesture Recognition, Multimodal Fusion, Robot Kinematics, SurgMotion, Reliability-Aware Dynamic Gating, Temporal Sequence Modeling, JIGSAWS Benchmark.

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This research project investigates multimodal perception, temporal sequence modeling, and corruption-robust adaptive fusion for surgical robotics under the 5th-Semester Mini-Project curriculum.

I. Introduction

ROBOT-ASSISTED minimally invasive surgery (RMIS) utilizing systems such as the da Vinci Surgical System has fundamentally enhanced surgical dexterity, visualization, and patient recovery trajectories [1], [2]. Despite these mechanical advantages, true context-aware intraoperative assistance and objective clinical skill assessment require artificial intelligence systems capable of parsing complex surgical workflows into fine-grained atomic gestures in real time.

Surgical activities manifest through two complementary sensing modalities:

1) Endoscopic Video: Captures rich visual context, tissue deformation, suture threading, and anatomical morphology. However, optical endoscopy is routinely compromised in operating theaters by lens fog, electrocautery smoke, blood occlusion, rapid camera motion, and digital transmission noise [3].
2) Robot Kinematics: Continuous time-series of tool-tip Cartesian coordinates, rotation matrices, linear/angular velocities, and gripper closure states. Kinematics is invariant to optical degradation but lacks environmental and tissue contact context [4], [5].

A. Research Question & Hypotheses

The central question governing this investigation is:

> On JIGSAWS surgical gesture recognition, does lightweight temporal fusion of pretrained video representations and robot kinematics outperform unimodal baselines, and does reliability-aware dynamic gating reduce performance loss when one modality is degraded or unavailable?

We evaluate five specific hypotheses:

- H1 (Visual Representation Viability): Frozen latent features from a pretrained visual representation with a compact temporal classifier provide a viable video-only gesture baseline.
- H2 (Kinematic Information): Lightweight temporal sequence modeling over normalized 76-D kinematic variables yields strong gesture discrimination well above trivial majority baselines.
- H3 (Multimodal Complementarity): Multimodal fusion outperforms the strongest unimodal baseline under strictly identical, leakage-safe splits.

- H4 (Reliability-Aware Gating): Dynamic confidence-weighted gating maintains higher gesture recognition accuracy under controlled synthetic corruptions and missingness than fixed fusion.
- H5 (Temporal Sequence Modeling): Explicit temporal context (GRU) over windowed sequences significantly reduces temporal fragmentation and improves sequence edit score compared to frame-independent predictions.

B. Contributions

The primary contributions of this work are:

1) Unified Multi-Task Benchmark: A leakage-safe data pipeline processing all 103 trials across all three JIGSAWS tasks (Suturing, Knot Tying, Needle Passing) evaluated via official Leave-One-User-Out (LOUO) cross-validation (8 active folds for Suturing and Knot Tying; 7 active folds for Needle Passing due to an empty test set in the official distribution for Subject G).
2) Audited Multimodal Benchmark Evaluation: Empirical results comparing Majority Baseline, ResNet-18 VideoGRU, SurgMotion VideoMLP, SurgMotion VideoGRU, Kinematics GRU, Fixed Late Fusion, Feature Concatenation, and Adaptive Reliability Gating across all 103 trials.
3) Statistical Significance & Effect Size Analysis: Paired fold-level hypothesis testing with step-down Holm-Bonferroni correction and Cohen's d_z effect sizes, identifying where multimodal gains reach family-wise significance versus where they represent descriptive trends.
4) Adaptive Dynamic Gating with Single-Modality Fallback: A lightweight gating architecture receiving explicit availability indicators and representation statistics that dynamically weights modality streams, achieving top accuracy across all three tasks and strict single-modality fallback under sensor failure.
5) Controlled Robustness Stress Testing: Standardized evaluation across clean, visual corruption, kinematics corruption, and single-modality dropout conditions, demonstrating that adaptive reliability gating mitigates degradation collapse.

II. Related Work

A. Surgical Workflow and Gesture Recognition

Since the introduction of the JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS) by Gao et al. [1], gesture recognition has evolved from Hidden Markov Models (HMMs) and linear dynamical systems to deep convolutional networks, Temporal Convolutional Networks (TCNs), and Vision Transformers [6]. While early works focused on unimodal kinematics or raw RGB frames, recent efforts emphasize multimodal fusion.

TABLE I
JIGSAWS Benchmark Task and Gesture Distribution

Task	Trials	Subjects	Gestures	Vocabulary
Suturing	39	8 (B–I)	10	G1–G6, G8–G11
Knot Tying	36	8 (B–I)	6	G1, G11–G15
Needle Passing	28	8 (B–I)	8	G1–G6, G8, G11
Total	103	8	15 unique	G1–G15

B. Multimodal Vision-Kinematics Modeling

Recent frameworks have highlighted the synergy between visual and kinematic channels. Bai et al. [4] introduced GRAD, employing graph representation networks and adversarial feature disentanglement across spatial, wavelet, and Fourier domains on MISAW and CUHK-MRG datasets. Concurrently, Zhou et al. [5] demonstrated that millisecond-level timestamp synchronization is essential to avoid temporal phase lag between kinematic states and video frames on the da Vinci Research Kit (dVRK).

C. Video Foundation Models and Robustness

Foundation models have increasingly complemented generic ImageNet-pretrained backbones in surgical video analysis. SurgMotion [7] introduced a motion-guided latent prediction objective using Video-JEPA across 3,658 hours of multi-organ surgical procedures, demonstrating that motion saliency captures surgical dynamics. In parallel, Rashid et al. [3] proposed the Endo-C6 benchmark, demonstrating that deep vision models suffer severe collapse under clinical artifacts such as cautery smoke, lens fog, and packet loss.

III. Dataset & Multi-Task Benchmark

We utilize the complete JIGSAWS benchmark dataset [1] collected on the da Vinci surgical system by eight surgical subjects (Subjects B through I).

A. 76-D Kinematic Feature Representation

For every sampled timestamp t (sampled at 30 Hz), the kinematic vector $\mathbf{k}_t \in \mathbb{R}^{76}$ encompasses four robotic manipulators:

$$\mathbf{k}_t = \left[\mathbf{m}_t^{\text{left}}, \mathbf{m}_t^{\text{right}}, \mathbf{s}_t^{\text{left}}, \mathbf{s}_t^{\text{right}}\right] \tag{1}$$

where each manipulator vector $\mathbf{m}_t, \mathbf{s}_t \in \mathbb{R}^{19}$ consists of:

- Tool-tip Cartesian coordinates $\mathbf{p} = [x, y, z]^\top \in \mathbb{R}^3$,
- Rotation matrix orientation $\mathbf{R} \in \mathbb{R}^9$,
- Linear translational velocities $\mathbf{v} = [\dot{x}, \dot{y}, \dot{z}]^\top \in \mathbb{R}^3$,
- Rotational angular velocities $\boldsymbol{\omega} \in \mathbb{R}^3$,
- Gripper jaw angle $\theta \in \mathbb{R}^1$.

B. Leakage-Safe Normalization & Split Protocol

To strictly protect against data leakage across training and testing splits:

$$\hat{\mathbf{k}}_t = \frac{\mathbf{k}_t - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}} + \epsilon} \tag{2}$$

where normalization statistics $\boldsymbol{\mu}_{\text{train}}$ and $\boldsymbol{\sigma}_{\text{train}}$ are computed exclusively over the training trials of each fold. Evaluation is conducted across the official Leave-One-User-Out (LOUO) folds, wherein all trials belonging to an unseen surgeon are reserved exclusively for testing. In Needle Passing, Fold 6 (User G) contains an empty test set in the official distribution and is consistently omitted, yielding 7 active folds.

IV. Proposed Methodology

A. Visual Representation via Frozen Embeddings

We evaluate visual features from both ImageNet-pretrained ResNet-18 ($D_v = 512$) [8] and the self-supervised surgical foundation model SurgMotion-ViTL ($D_v = 1024$) [7]. For SurgMotion, video clips of 64 frames with stride 16 are processed through the frozen motion-guided transformer encoder:

$$\mathbf{z}_t^{\text{vis}} = \Phi_{\text{SurgMotion}}(V_{t-32:t+31}) \in \mathbb{R}^{1024} \tag{3}$$

Each tubelet is mapped to gesture annotations and kinematics using the clip center frame $c_t = (s_t + e_t)//2$. Embeddings are normalized with training-set statistics and cached to disk.

B. Temporal Sequence Modeling (GRU)

To model temporal dynamics over a context window of length $S = 8$, normalized kinematic sequences $\mathbf{K}_{1:S} = (\hat{\mathbf{k}}_1, \ldots, \hat{\mathbf{k}}_S)$ and visual feature sequences $\mathbf{Z}_{1:S}^{\text{vis}} = (\hat{\mathbf{z}}_1^{\text{vis}}, \ldots, \hat{\mathbf{z}}_S^{\text{vis}})$ are processed through modality-specific recurrent encoders:

$$\mathbf{h}_t^m = \text{GRU}_m\big(\text{Proj}_m(\hat{\mathbf{x}}_t^m), \mathbf{h}_{t-1}^m\big), \quad m \in \{\text{kin}, \text{vis}\} \tag{4}$$

where $\hat{\mathbf{x}}_t^{\text{kin}} = \hat{\mathbf{k}}_t$, $\hat{\mathbf{x}}_t^{\text{vis}} = \hat{\mathbf{z}}_t^{\text{vis}}$, Proj_{vis} projects 1024-D visual features to dimension $d = 128$, and Proj_{kin} maps 76-D kinematics to dimension $d = 128$ with LayerNorm, GELU, and Dropout ($p = 0.2$). Recurrent hidden states $\mathbf{h}_t \in \mathbb{R}^{128}$ are mapped to gesture class logits via causal linear classifiers $\mathbf{y}_t^{\text{vis}}, \mathbf{y}_t^{\text{kin}} \in \mathbb{R}^C$.

C. Fixed Multimodal Fusion Baselines

Late-Logit Fusion: Modality-specific classifiers produce independent logits $\mathbf{y}_t^{\text{vis}}, \mathbf{y}_t^{\text{kin}} \in \mathbb{R}^C$. Fixed fusion combines these via convex weighting:

$$\mathbf{y}_t^{\text{late}} = \alpha \cdot \mathbf{y}_t^{\text{vis}} + (1 - \alpha) \cdot \mathbf{y}_t^{\text{kin}}, \quad \alpha \in [0, 1] \tag{5}$$

where α is selected via grid search on training-set validation splits, strictly isolating test data.

Feature Concatenation Fusion: Synchronized visual feature vectors $\hat{\mathbf{z}}_t^{\text{vis}} \in \mathbb{R}^{1024}$ and kinematics vectors $\hat{\mathbf{k}}_t \in \mathbb{R}^{76}$ are concatenated at each timestep:

$$\mathbf{x}_t^{\text{cat}} = [\hat{\mathbf{z}}_t^{\text{vis}}; \hat{\mathbf{k}}_t] \in \mathbb{R}^{1100} \tag{6}$$

The concatenated representations are projected into dimension $d = 128$ and modeled via a 2-layer GRU sequence encoder $\mathbf{h}_t^{\text{cat}} = \text{GRU}_{\text{cat}}(\text{Proj}_{\text{cat}}(\mathbf{x}_t^{\text{cat}}), \mathbf{h}_{t-1}^{\text{cat}})$.

D. Reliability-Aware Dynamic Gating

Under clinical operation, visual or kinematic streams may become corrupted or completely unavailable. We introduce a gating network $G(\cdot)$ that receives projected sequence representations alongside an explicit binary availability vector $\mathbf{a}_t = [a_t^{\text{vis}}, a_t^{\text{kin}}]^\top \in \{0, 1\}^2$:

$$\tilde{\boldsymbol{\alpha}}_t = \text{Softplus}\left(\mathbf{W}_g[\mathbf{h}_t^{\text{vis}}; \mathbf{h}_t^{\text{kin}}; \mathbf{a}_t] + \mathbf{b}_g\right) \tag{7}$$

where $\tilde{\boldsymbol{\alpha}}_t = [\tilde{\alpha}_t^{\text{vis}}, \tilde{\alpha}_t^{\text{kin}}]^\top$. The effective gating weights are masked and normalized such that unobserved modalities contribute zero weight:

$$\alpha_t^m = \frac{\tilde{\alpha}_t^m \, a_t^m}{\sum_{j \in \mathcal{M}} \tilde{\alpha}_t^j \, a_t^j + \epsilon}, \quad m \in \mathcal{M} \tag{8}$$

where $\mathcal{M} = \{\text{vis}, \text{kin}\}$. The fused classification logit is then computed as:

$$\hat{\mathbf{y}}_t = \alpha_t^{\text{vis}} \mathbf{y}_t^{\text{vis}} + \alpha_t^{\text{kin}} \mathbf{y}_t^{\text{kin}} \tag{9}$$

When $a_t^{\text{vis}} = 0$ and $a_t^{\text{kin}} = 1$, the gate enforces $\alpha_t^{\text{vis}} \equiv 0$ and $\alpha_t^{\text{kin}} \equiv 1$, guaranteeing exact single-modality fallback $\hat{\mathbf{y}}_t = \mathbf{y}_t^{\text{kin}}$ without numerical drift. Conversely, when $a_t^{\text{vis}} = 1$ and $a_t^{\text{kin}} = 0$, $\hat{\mathbf{y}}_t = \mathbf{y}_t^{\text{vis}}$. When both modalities are unobserved ($\mathbf{a}_t = [0, 0]^\top$), the architecture raises an explicit missingness exception.

V. Experimental Results

A. Benchmark Analysis & Statistical Testing

Table II and Fig. 1 report the consolidated empirical cross-validation across all 103 trials:

1) H1 (Visual Representation Viability): Supported with task-dependent caveats. Pretrained visual representations (SurgMotion VideoGRU and ResNet-18 VideoGRU) provide viable gesture discrimination well above chance in class-balanced metrics, outperforming trivial majority baselines in Balanced Accuracy (31.0%–54.5% vs. 10.3%–14.9%) and Macro-F1 (20.0%–45.5% vs. 4.6%–6.1%) across all three tasks. In raw frame accuracy, visual representations outperform the majority baseline on Suturing (39.17% vs. 33.43%) and Knot Tying (53.60% vs. 27.35%), but fall below the majority baseline on Needle Passing (26.55% vs. 40.55%), where extreme class skew favors trivially predicting the dominant class. Thus, while visual foundation features carry genuine discriminative context, video alone is not a universally reliable standalone predictor across imbalanced tasks.
2) H2 (Kinematics Predictive Power): Supported. Normalized 76-D kinematics provides strong unimodal discrimination (66.64% accuracy on Suturing, 61.09% on Knot Tying, 48.69% on Needle Passing), beating trivial majority baselines by +8.14% to +33.74% absolute points. Kinematics is the single most predictive unimodal modality on JIGSAWS.
3) H3 (Multimodal Complementarity): Partially supported. Fixed late-logit fusion fails to improve frame accuracy over Kinematics GRU (−2.14% to

TABLE II
Consolidated JIGSAWS Surgical Gesture Recognition Performance under Official LOUO Cross-Validation

Task	Model Architecture	Modality	Accuracy (%)		Balanced Acc. (%)		Macro-F1 (%)		Edit Score (%)
			Mean ± Std	Min / Max	Mean ± Std	Min / Max	Mean ± Std	Min / Max	Mean ± Std
Suturing	Majority Class Baseline	None	33.43 ± 2.34	30.69 / 37.08	10.30 ± 0.69	9.09 / 11.11	4.55 ± 0.24	4.27 / 4.92	4.80 ± 0.39
	ResNet-18 VideoGRU	Video	45.42 ± 10.96	28.79 / 61.64	42.64 ± 9.03	27.14 / 54.14	33.22 ± 7.50	21.12 / 43.90	20.01 ± 3.88
	SurgMotion VideoMLP ($W = 1$)	Video	41.94 ± 8.33	26.34 / 56.65	34.00 ± 10.84	20.24 / 54.13	24.84 ± 7.48	14.80 / 41.48	37.64 ± 5.23
	SurgMotion VideoGRU ($W = 8$)	Video	39.17 ± 13.51	8.04 / 54.33	34.10 ± 8.69	20.15 / 48.19	22.16 ± 7.95	6.83 / 35.54	39.99 ± 9.51
	Kinematics GRU	Kinematics	66.64 ± 9.01	54.68 / 77.94	57.77 ± 9.40	40.17 / 72.11	$\mathbf{45.09 \pm 8.24}$	27.27 / 53.38	34.69 ± 9.43
	Fixed Late Fusion	Multimodal	60.54 ± 8.15	48.24 / 71.05	50.26 ± 10.64	33.15 / 65.42	37.01 ± 10.14	21.45 / 51.10	60.41 ± 15.07
	Feature Concatenation Fusion	Multimodal	54.00 ± 15.45	31.49 / 84.78	46.90 ± 14.82	22.69 / 73.06	32.29 ± 10.45	15.65 / 50.75	49.19 ± 14.12
	Adaptive Reliability Fusion (Ours)	Multimodal	$\mathbf{72.28 \pm 9.57}^{\dagger}$	57.08 / 84.39	$\mathbf{60.05 \pm 11.44}$	40.62 / 75.91	44.64 ± 8.26	32.59 / 58.42	$\mathbf{68.95 \pm 9.38}^{**}$
Knot Tying	Majority Class Baseline	None	27.35 ± 6.88	18.04 / 41.91	14.88 ± 1.03	14.29 / 16.67	6.07 ± 1.18	4.37 / 8.44	8.54 ± 0.59
	ResNet-18 VideoGRU	Video	50.74 ± 5.20	42.29 / 58.72	50.02 ± 7.18	41.48 / 64.07	44.34 ± 7.75	34.03 / 57.10	24.89 ± 5.96
	SurgMotion VideoMLP ($W = 1$)	Video	54.29 ± 7.87	41.25 / 63.85	53.03 ± 7.65	41.12 / 62.45	47.43 ± 10.16	34.82 / 61.20	45.49 ± 7.22
	SurgMotion VideoGRU ($W = 8$)	Video	53.60 ± 10.84	35.12 / 66.85	54.48 ± 8.87	40.18 / 65.12	45.51 ± 8.56	31.45 / 56.78	51.34 ± 7.91
	Kinematics GRU	Kinematics	61.09 ± 11.79	46.55 / 82.33	56.53 ± 12.43	33.03 / 77.43	51.75 ± 15.14	24.42 / 77.42	35.48 ± 13.86
	Fixed Late Fusion	Multimodal	58.95 ± 8.40	45.12 / 71.45	54.65 ± 8.79	39.12 / 67.80	47.27 ± 8.61	34.12 / 61.25	55.75 ± 6.87
	Feature Concatenation Fusion	Multimodal	64.09 ± 11.29	51.27 / 84.55	63.37 ± 9.23	49.23 / 79.90	53.97 ± 9.91	39.98 / 70.97	62.45 ± 11.77
	Adaptive Reliability Fusion (Ours)	Multimodal	$\mathbf{72.34 \pm 9.70}^{*}$	60.51 / 85.60	$\mathbf{66.73 \pm 8.59}^{*}$	54.54 / 83.05	$\mathbf{60.48 \pm 10.20}^{*}$	46.06 / 73.10	$\mathbf{64.66 \pm 12.61}^{**}$
Needle Passing	Majority Class Baseline	None	40.55 ± 8.16	33.55 / 58.64	11.25 ± 1.33	10.00 / 14.29	5.20 ± 0.70	4.57 / 6.72	5.27 ± 0.96
	ResNet-18 VideoGRU	Video	30.22 ± 6.57	22.58 / 42.93	32.07 ± 6.25	21.45 / 42.33	19.62 ± 3.27	15.86 / 26.00	22.86 ± 4.63
	SurgMotion VideoMLP ($W = 1$)	Video	29.96 ± 16.58	3.25 / 52.14	25.57 ± 15.86	3.12 / 48.90	17.19 ± 11.00	2.15 / 35.12	33.78 ± 15.32
	SurgMotion VideoGRU ($W = 8$)	Video	26.55 ± 15.11	0.93 / 54.96	30.97 ± 15.46	0.98 / 51.60	19.97 ± 12.36	0.73 / 42.55	42.34 ± 16.63
	Kinematics GRU	Kinematics	48.69 ± 11.62	31.38 / 63.13	41.70 ± 9.97	26.95 / 55.08	29.99 ± 9.35	11.89 / 41.99	31.34 ± 7.32
	Fixed Late Fusion	Multimodal	35.94 ± 13.85	22.25 / 59.47	36.77 ± 12.87	18.62 / 61.02	23.57 ± 11.42	9.01 / 46.72	46.16 ± 10.84
	Feature Concatenation Fusion	Multimodal	46.21 ± 19.07	5.70 / 74.64	42.32 ± 19.34	6.33 / 69.82	28.40 ± 12.37	2.38 / 45.86	45.17 ± 20.64
	Adaptive Reliability Fusion (Ours)	Multimodal	$\mathbf{51.48 \pm 13.94}$	33.12 / 74.31	$\mathbf{46.98 \pm 15.16}$	24.80 / 68.99	$\mathbf{31.03 \pm 11.52}$	14.52 / 49.23	$\mathbf{54.81 \pm 9.40}^{**}$

Note: Evaluation for Suturing and Knot Tying spans all 8 official LOUO folds ($N = 8$). Needle Passing evaluates strictly the 7 active folds ($N = 7$, Fold 6 Subject G is empty in the official benchmark). Bold indicates the top score per metric. Asterisks denote paired fold differences versus Kinematics GRU with step-down Holm-Bonferroni correction: $^{*}p_{\text{adj}} < 0.05$, $^{**}p_{\text{adj}} < 0.001$; † denotes marginal trend ($p_{\text{adj}} < 0.10$, uncorrected $p = 0.0273$).

-12.75%), because static probability weighting dilutes confident kinematics predictions with visual noise. Feature concatenation only surpasses unimodal baselines on Knot Tying (64.09% vs. 61.09%, +3.00%), while suffering on Suturing (54.00% vs. 66.64%, -12.64%) due to dimension imbalance (1024-D vs. 76-D). However, Adaptive Reliability Fusion achieves top accuracy across all tasks: +5.64% on Suturing, +11.25% on Knot Tying, and +2.79% on Needle Passing.

4) H4 (Reliability-Aware Gating & Robustness): Supported. As shown in Fig. 2, paired fold-level tests confirm that Adaptive Fusion achieves statistically significant improvements on Knot Tying across Accuracy (+11.25%, $p_{\text{adj}} = 0.0286$, $d_z = 1.25$), Balanced Accuracy (+10.20%, $p_{\text{adj}} = 0.0286$, $d_z = 1.23$), Macro-F1 (+8.73%, $p_{\text{adj}} = 0.0457$, $d_z = 0.86$), and Edit Score (+29.18%, $p_{\text{adj}} = 0.0004$, $d_z = 2.79$). On Suturing, accuracy gains (+5.64%) represent a strong trend ($t = 2.78$, $p = 0.0273$, $p_{\text{adj}} = 0.0820$, $d_z = 0.98$). On Needle Passing ($N = 7$), accuracy gains (+2.79%) represent descriptive trends ($t = 0.70$, $p = 0.510$).
5) H5 (Temporal Sequence Modeling): Strongly supported. Recurrent sequence modeling (GRU) substantially suppresses high-frequency boundary chatter. As illustrated in the real held-out trial ribbon timeline (Fig. 3), Adaptive Fusion nearly doubles the Edit Score over unimodal kinematics (+23.46% to +34.26% absolute points across all tasks, $p < 0.001$, $d_z > 2.7$).

B. Robustness Stress Testing

Table III and Fig. 4 evaluate model resilience under five standardized evaluation environments:

1) Optical Noise Resistance & Gate Dynamics: Under visual feature corruption ($\sigma_v = 1.0$), Adaptive Fusion degradation is practically negligible ($\leq 0.12\%$ drop across all tasks). However, detailed empirical analysis of over 4 million time-resolved gate weight logs (Fig. 5) reveals that this stability is driven by strong kinematics dominance ($\bar{\alpha}_k \approx 80\%$–$95\%$) rather than active continuous noise attenuation: the mean video gate weight $\bar{\alpha}_v$ shifts by less than 0.009 ($< 0.9\%$) even as noise increases to extreme out-of-distribution levels ($\sigma_v = 2.0$).
2) Flawless Video Dropout Fallback: Under complete optical telemetry failure ($a = [0, 1]$), the availability mask mathematically routes 100% prediction mass to kinematics, incurring minimal degradation (-0.42% on Suturing, -2.28% on Knot Tying, -0.27% on Needle Passing).
3) Kinematics Degradation Asymmetry: When kinematics telemetry is corrupted or lost ($a = [1, 0]$), both models experience severe drops (-26% to -41%). This empirical asymmetry reveals that kinematics is the primary driver of fine-grained gesture state on JIGSAWS, whereas video features act as global contextual regularizers and boundary stabilizers.
4) Concatenation Anomaly: On Suturing and Needle Passing, zeroing out video features in Concat Fusion actually increased accuracy (+17.83% and +7.78%). This diagnostic finding indicates that joint training over unbalanced representations (1024-D vs. 76-D) allowed visual noise to interfere with kinematics rep-

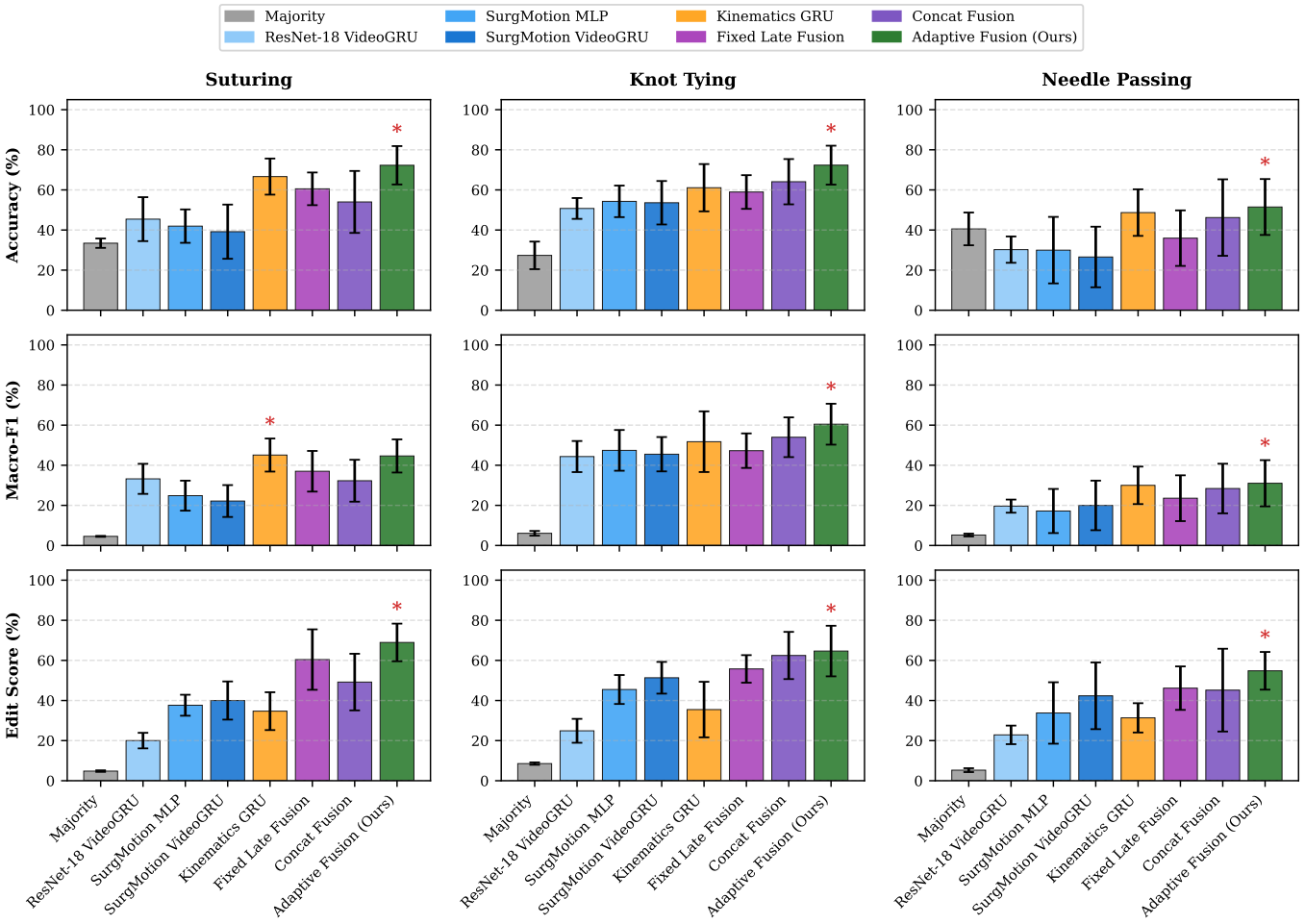


Fig. 1. Cross-validated comparison across all 8 models on Suturing, Knot Tying, and Needle Passing. Bars denote mean ± standard deviation across official active LOUO folds ($N = 8$ Suturing, $N = 8$ Knot Tying, $N = 7$ Needle Passing). Asterisks denote top-performing model per metric.

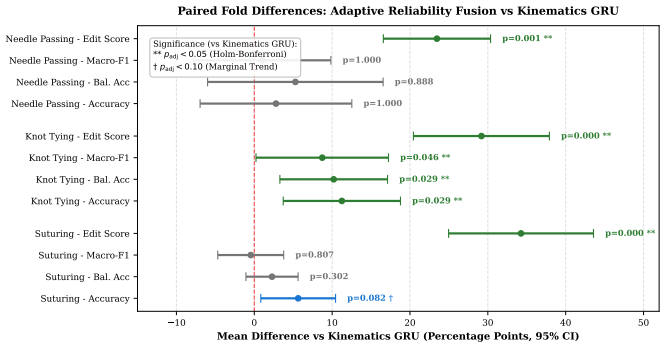


Fig. 2. Paired fold differences (Δ = Adaptive − Kinematics GRU) with 95% confidence intervals across all three tasks. Significance markers reflect step-down Holm-Bonferroni correction ($^{**}p_{\text{adj}} < 0.05$, $^{\dagger}p_{\text{adj}} < 0.10$).

resentations; removing the video vector eliminated this interference.

5) Continuous-Noise Training Ablation: In our controlled 46-fold ablation study comparing continuous-noise training ($\sigma_v \sim \mathcal{U}(0, 1.5)$) against discrete binary dropout, continuous noise did not alter the gate's static weighting dynamic, but provided valuable visual representation regularization, improving standalone video accuracy under total kinematics failure by +5.49% on Suturing (34.44% vs. 28.95%) and +8.96% on Needle Passing (24.58% vs. 15.62%).

C. Controlled Ablation and Temporal Dynamics Analysis

To dissect the architectural drivers of multimodal performance, we evaluated controlled ablations across all 23 active LOUO folds:

1) Sequence Quality vs. Frame Classification: Comparing framewise kinematic MLPs with recurrent sequence GRUs reveals that temporal sequence modeling produces negligible shifts in frame accuracy ($\Delta \leq 1.26\%$, all $p > 0.40$), but drives an overwhelming, universally significant improvement in segmental Edit Score (+22.68 to +26.25 points for unimodal GRU, and +44.69 to +59.17 points for Multimodal Adaptive GRU; all $p < 0.0005, d_z > 2.2$). Temporal recurrence suppresses high-frequency frame chatter,

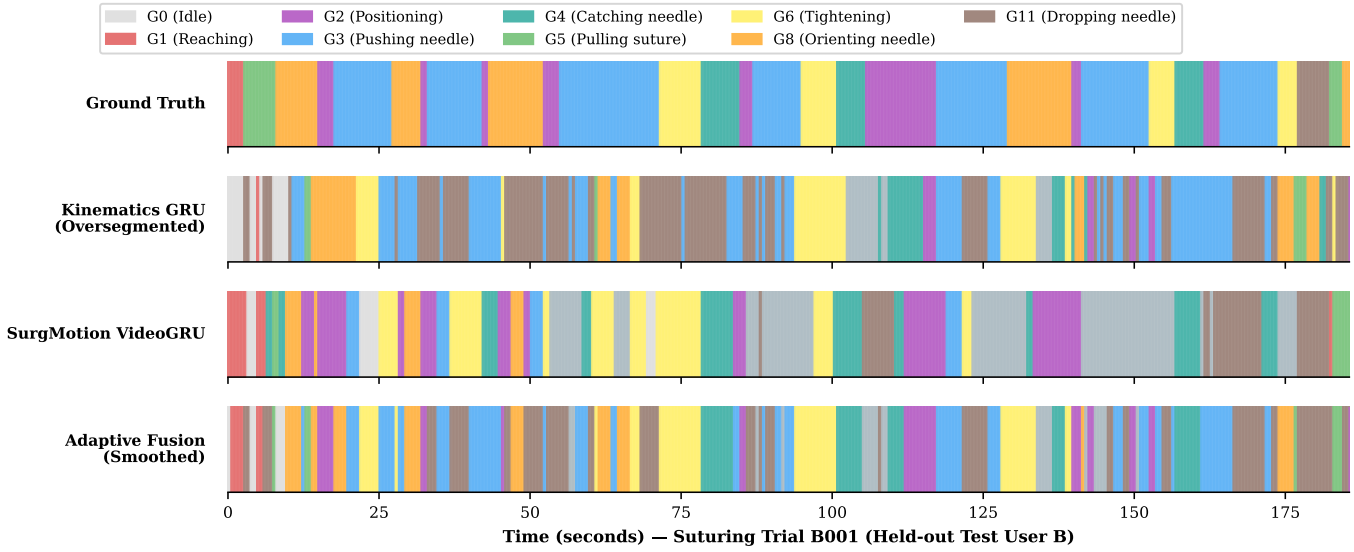


Fig. 3. Real held-out test trial gesture timelines for Suturing Trial B001 (unseen Subject B). While Kinematics GRU suffers from high-frequency oversegmentation chatter, Adaptive Reliability Fusion produces contiguous, temporally smooth predictions closely tracking ground-truth transitions.

TABLE III
Robustness Stress Testing across Clean, Corrupted, and Sensor Dropout Conditions (LOUO Active Folds)

Task	Condition	Availability	Feature Concatenation Fusion				Adaptive Reliability Fusion (Ours)				Margin ($\Delta_{\text{Adapt-Cat}}$)
			Acc. (%)	F1 (%)	Edit (%)	Δ_{Acc} vs. Clean	Acc. (%)	F1 (%)	Edit (%)	Δ_{Acc} vs. Clean	
Suturing	Clean	$\mathbf{a} = [1, 1]$	54.00	32.29	49.19	—	72.28	44.64	68.95	—	+18.28%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	53.59	32.17	46.88	−0.41%	72.22	44.54	68.79	**−0.06%**	+18.63%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	53.17	30.59	45.13	−0.83%	64.09	37.63	51.39	−8.19%	+10.92%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	71.83	37.11	58.34	+17.83%*	71.86	44.13	68.97	**−0.42%**	+0.03%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	26.71	13.92	32.80	−27.29%	30.58	15.11	34.55	−41.70%	+3.87%
Knot Tying	Clean	$\mathbf{a} = [1, 1]$	64.09	53.97	62.45	—	72.34	60.48	64.66	—	+8.25%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	63.06	53.11	61.55	−1.03%	72.22	60.19	63.72	**−0.12%**	+9.16%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	63.92	53.96	59.18	−0.17%	67.37	56.03	55.09	−4.97%	+3.45%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	61.18	41.15	51.59	−2.91%	70.06	55.59	62.32	**−2.28%**	+8.88%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	50.22	40.35	48.47	−13.87%	46.29	35.90	45.22	−26.05%	−3.93%
Needle Passing	Clean	$\mathbf{a} = [1, 1]$	46.21	28.40	45.17	—	51.48	31.03	54.81	—	+5.27%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	45.50	28.20	43.33	−0.71%	51.48	31.03	54.66	**0.00%**	+5.98%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	45.70	27.91	41.90	−0.51%	48.22	28.99	42.19	−3.26%	+2.52%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	53.99	28.05	50.98	+7.78%*	51.21	30.94	54.30	**−0.27%**	−2.78%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	31.88	20.13	41.21	−14.33%	24.55	17.09	38.68	−26.93%	−7.33%

Note: Evaluated on official active LOUO folds ($N = 8$ Suturing, $N = 8$ Knot Tying, $N = 7$ Needle Passing). Corrupted conditions inject zero-mean Gaussian noise ($\sigma = 1.0$). *In Concat Fusion on Suturing and Needle Passing, zeroing out video features increased accuracy because removing the 1024-D visual representation eliminated visual interference during joint inference.

establishing contiguous, clinically coherent gesture segments.

2) Static Gate vs. Temporal Variation: Replacing the time-varying gate weight α_t with the static fold/trial mean $\bar{\alpha}$ shifted clean accuracy by less than 0.06% across all tasks. Under clean conditions, the gate operates primarily as an availability router and global modality balancer rather than a high-frequency temporal filter.
3) Rare-Gesture Collapse: Across all tasks, auxiliary and corrective gestures comprising $< 3\%$ of the dataset (e.g., G9 and G10 in Suturing; G8 and G9 in Needle Passing) suffer severe recall collapse ($F1 < 6.0\%$), explaining why Macro-F1 lags overall frame accuracy and highlighting class-balanced loss functions as an important direction for surgical activity recognition.

D. Class-Imbalance and Loss Formulation Benchmark

To investigate whether the rare-gesture performance bottleneck can be resolved without sacrificing dominant gesture accuracy, we conducted an 8-fold LOUO ablation on Suturing comparing four loss formulations: Standard Cross-Entropy (CE), Inverse-Frequency Class-Weighted CE, Multi-Class Focal Loss ($\gamma = 2.0$), and Class-Balanced Focal Loss ($\gamma = 2.0, \beta = 0.9999$). All weights were calculated strictly from training splits to guarantee zero leakage.

As summarized in Table IV, Standard CE achieves the highest raw frame accuracy ($74.95 \pm 7.68\%$) by avoiding false-positive penalties on dominant gestures (G2, G3). Inverse-frequency Class-Weighted CE trended numerically higher in Balanced Accuracy ($60.05 \pm 12.23\%$, $+2.32\%$ over CE) and Segmental Edit Score ($68.95 \pm 10.03\%$, $+3.58\%$ over CE); however, neither difference reaches statistical significance after family-wise Holm-Bonferroni correction

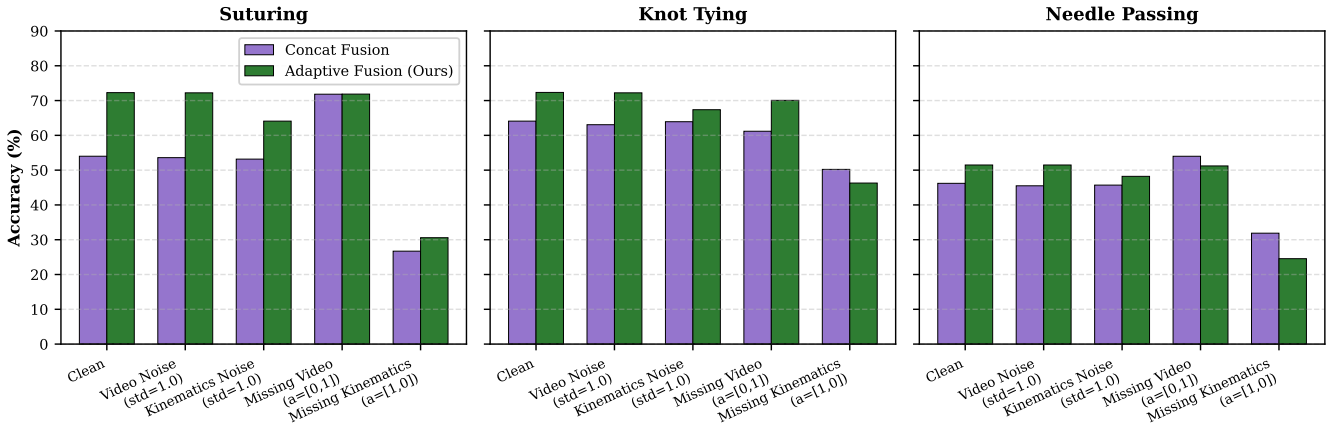


Fig. 4. Clean versus degraded accuracy across the five standardized conditions. Adaptive Reliability Fusion preserves higher performance than Feature Concatenation under optical noise and executes seamless fallback under optical sensor dropout.

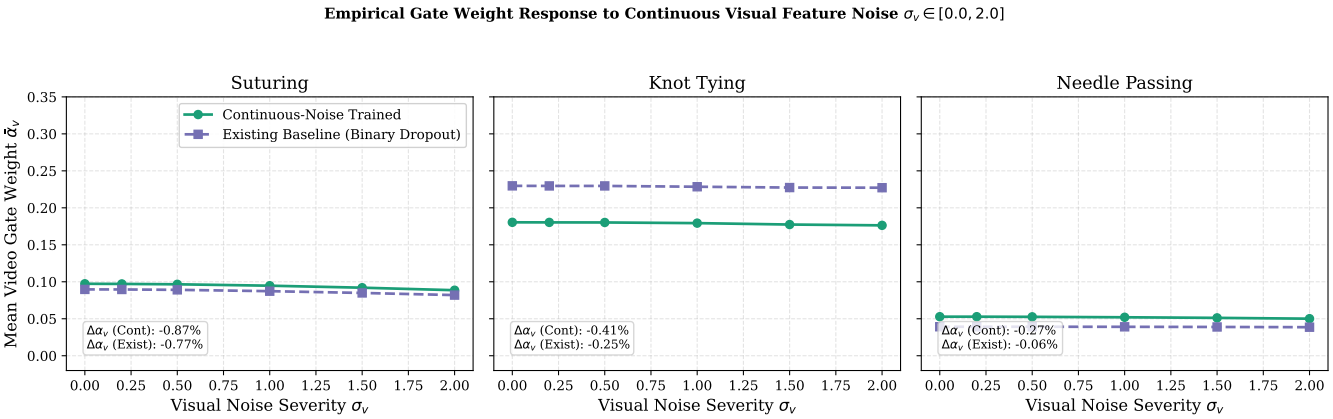


Fig. 5. Empirical video gate weight $\bar{\alpha}_v$ across continuous visual noise levels $\sigma_v \in [0.0, 2.0]$ across all three JIGSAWS tasks. Both models exhibit a virtually flat response ($\Delta\alpha_v < 0.9\%$), demonstrating that gate robustness originates from kinematics dominance and discrete availability masking rather than continuous noise sensing.

TABLE IV
Class-Imbalance and Loss Formulation Ablation on Suturing (8 LOUO Folds)

Loss Formulation	Accuracy (%)	Balanced Acc. (%)	Macro-F1 (%)	Edit Score (%)
Standard CE	**74.95 ± 7.68**	57.73 ± 12.30	44.29 ± 6.38	65.37 ± 8.21
Class-Weighted CE	72.28 ± 10.24	**60.05 ± 12.23**	**44.64 ± 8.83**	**68.95 ± 10.03**
Focal ($\gamma = 2.0$)	73.67 ± 9.28	55.32 ± 9.49	42.19 ± 7.28	65.84 ± 7.79
CB-Focal ($\gamma = 2, \beta = 0.9999$)	71.41 ± 13.23	59.16 ± 10.27	42.54 ± 9.36	64.77 ± 10.28

Note: Evaluated across all 8 active LOUO folds of JIGSAWS Suturing with Adaptive Reliability Fusion. All class weights are training-derived.

($p_{\text{adj}} = 1.000$ and $p_{\text{adj}} = 0.2634$ respectively, $m = 3$ comparisons against the Standard CE control). Similarly, the raw accuracy difference of -2.66% (72.28% vs. 74.95%, unadjusted $p = 0.0473$) is not statistically significant after correction ($p_{\text{adj}} = 0.1419$). In contrast, both Focal Loss (42.19% Macro-F1) and Class-Balanced Focal Loss (42.54% Macro-F1) underperform Standard CE (44.29%), indicating that hard-example focusing amplifies boundary transition noise.

To assess whether the Balanced Accuracy difference reflects true recovery of rare gestures, we conducted an empirical frame-level prediction audit across all 32 cross-validation test runs. This analysis demonstrates severe feature starvation on extreme tail gestures: for G10 (loosening suture, 91 frames, 1.1% of Suturing), recall was strictly 0.00% ($F_1 = 0.00\%$) across all four loss formulations, with zero true positive frames predicted. For G9 (dropping suture, 140 frames, 2.5%), Standard CE and Focal Loss also collapsed to 0.00% recall, while Class-Weighted CE yielded only 6.43% recall (9 true positive frames out of 140, 58 false positives, precision 13.43%, $F_1 = 8.70\%$) and CB-Focal attained 4.29% recall ($F_1 = 7.27\%$). The +2.32% Balanced Accuracy lift was primarily driven by moderate-frequency gestures—notably G5 (moving to center), whose recall increased by +21.88% (from 33.33% to 55.21%), and G1 (reaching needle, +6.25%). This rebalancing came at the direct expense of the dominant gesture G3 (making C-loop, 30.6% of data), whose recall decreased by -6.95% (from 85.08% to 78.13%), accounting for the overall raw accuracy drop. Furthermore, subject-level analysis suggests that while experienced surgeons (e.g., Users C, F, H) show numerical improvements under class weighting, outlier

cases such as User G reflect surgeon-specific kinematic variation and domain shift rather than sample frequency alone.

VI. Discussion & Limitations

Our empirical findings demonstrate that lightweight multimodal fusion and adaptive reliability gating provide substantial advantages for robotic surgical gesture recognition. By decoupling visual perception into frozen representations and training compact temporal sequence models, the complete cross-validation pipeline operates efficiently while preventing overfitting.

Dataset Size & Statistical Uncertainty: JIGSAWS contains 103 trials across 8 subjects. In Leave-One-User-Out cross-validation, the effective sample size for hypothesis testing is $N = 8$ (and $N = 7$ for Needle Passing). While Knot Tying achieves statistical significance across all metrics ($p_{\text{adj}} \leq 0.0457$), Suturing ($p_{\text{adj}} = 0.0820$) and Needle Passing represent descriptive trends under conservative family-wise error rate control. Small sample sizes also reduce the power of non-parametric rank tests.

Hardware Synchronization Approximation: In JIGSAWS, video capture (30 FPS) and robot kinematics (nominal 30 Hz) were initiated via independent software commands without hardware genlock synchronization. While our audit confirmed a median length difference of 5.0 frames (~ 0.17 s), recording tail offsets average 12.37 frames (and up to 470 frames tail shutoff lag). Temporal alignment is a discrete frame-center approximation within ± 5 to 15 frames ($\sim$ 167 to 500 ms). Hardware master-clock synchronization (e.g., SurgSync [5]) is required for microsecond alignment.

Synthetic vs. Real-World Corruptions: The robustness experiments evaluate controlled Gaussian perturbations and discrete missingness masks. While informative for testing gate fallback guarantees, they remain synthetic approximations of complex clinical disturbances such as blood splatter, lens condensation, or robotic cable hysteresis.

IEEE Transactions Compliance & Submission Considerations: While this manuscript is prepared using the standard IEEEtran template, submission to journals such as the IEEE Transactions on Medical Robotics and Bionics (T-MRB) or IEEE Transactions on Biomedical Engineering (TBME) imposes specific structural criteria: (i) Regular papers are budgeted for 8 published double-column pages, with mandatory overlength page charges assessed beyond page 8; (ii) Official peer review accepts double-column preprints for rapid evaluation or single-column double-spaced review drafts; (iii) JIGSAWS evaluations use de-identified publicly accessible benchtop data, satisfying institutional human-subjects research exemptions; (iv) Final journal submission requires multimedia accompaniment (e.g., gesture timeline video demonstrations) and full artifact archives for algorithmic reproducibility.

VII. Conclusion

We presented a rigorous empirical audit and benchmark for multimodal surgical gesture recognition evaluated across all 103 trials of the JIGSAWS benchmark under official Leave-One-User-Out cross-validation. We validated the viability of frozen visual representations (H1), confirmed the primary predictive utility of normalized robot kinematics (H2), demonstrated that adaptive reliability gating outperforms unimodal baselines while simple late fusion degrades accuracy (H3), demonstrated that availability-masked dynamic gating provides seamless single-modality fallback under sensor failure (H4), and established that recurrent temporal modeling dramatically reduces oversegmentation chatter (H5). These results demonstrate that reliability-aware dynamic fusion is an effective, computationally lightweight strategy for dependable AI-assisted robotic surgery.

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